

CS166 Lab 7 Assignment

Indexing

Exercise 1

```
INSERT 0 1
[dtape004@xe-10 cs166_lab7_updated]$ cs166_psql dtape004_lab7_DB < queries.sql
count
-----
 7854
(1 row)

sum
-----
1490350
 874230
(2 rows)

supplier_name
-----
Bob
Ted
Carol
Alice
Jerry
Mickey
(6 rows)

supplier_name
-----
(0 rows)

UPDATE 49995
DELETE 8456
[dtape004@xe-10 cs166_lab7_updated]$
```

Figure 1: queries.sql Results

Exercise 2

```
PROBLEMS OUTPUT TERMINAL PORTS SERIAL MONITOR bash - cs166_lab7_updated

DROP TABLE
CREATE TABLE
INSERT 0 1
INSERT 0 1
Query time without indexes
Query1: 7.415 ms
Query2: 17.372 ms
Query3: 71.716 ms
Query4: 7.449 ms
Query5: 106.012 ms
Query6: 13.287 ms
NOTICE: index "idx_nyc_on_hand" does not exist, skipping
NOTICE: index "idx_sfo_on_hand" does not exist, skipping
NOTICE: index "idx_nyc_color" does not exist, skipping
NOTICE: index "idx_sfo_color" does not exist, skipping
NOTICE: index "idx_nyc_supplier" does not exist, skipping
NOTICE: index "idx_sfo_supplier" does not exist, skipping
Query time with indexes
Query1: 0.747 ms
Query2: 14.786 ms
Query3: 25.617 ms
Query4: 8.066 ms
Query5: 282.704 ms
Query6: 12.914 ms
[dtape004@xe-10 cs166_lab7_updated]$
```

-- NORMAL -- Ln 5, Col 1 Spaces: 4 UT

Figure 2: Speed of Queries with and without Indexes

Query time without Indexes

	Query 1	Query 2	Query 3	Query 4	Query 5	Query 6
Trial 1	7.415	17.372	71.716	7.449	106.012	13.287
Trial 2	8.220	19.468	77.281	7.306	79.625	13.737
Trial 3	7.486	17.852	71.990	7.012	72.876	13.240
Trial 4	8.226	19.452	91.598	8.142	177.872	16.071
Trial 5	8.546	20.325	82.630	7.093	131.204	15.128
Average	7.979	18.894	79.043	7.400	113.518	14.293

Query time with Indexes

	Query 1	Query 2	Query 3	Query 4	Query 5	Query 6
Trial 1	0.747	14.786	25.617	8.066	282.704	12.914
Trial 2	0.734	14.024	25.283	7.013	200.641	12.627
Trial 3	0.793	14.713	24.745	6.961	261.942	12.240
Trial 4	0.752	16.708	29.511	7.283	317.941	14.558
Trial 5	0.787	16.291	28.204	7.323	230.764	14.580
Average	0.763	15.304	26.672	7.329	258.798	13.384